Cyber Shujaa

Security Analyst

ASSIGNMENT REPORT

DNS IN Detail

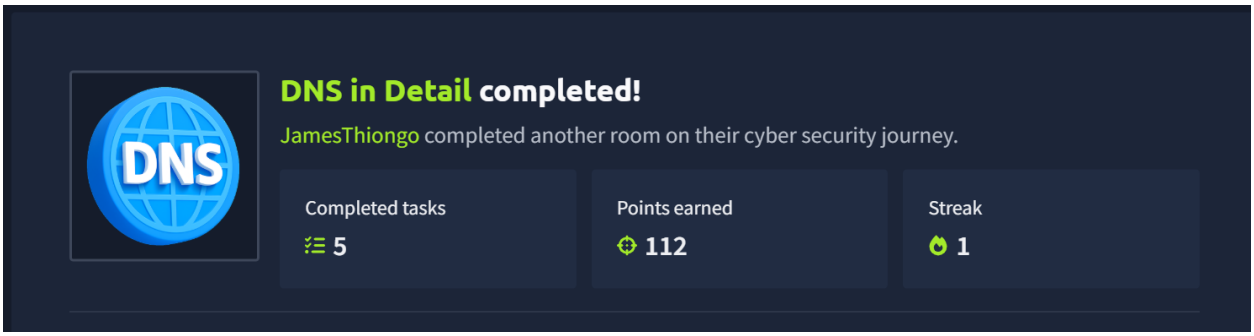
Full name	James Thiongo
Admission number	
Email	
Week and assignment	Week 4, Assignment 2: DNS in Detail
Platform	TryHackMe
Date submitted	05/10/2026

1. Module completion

Shareable link

https://tryhackme.com/room/dnsindetail?utm_campaign=social_share&utm_medium=social&utm_content=share-completed-room&utm_source=copy&sharerId=67c6e381a05df35cdc1f7b1c

Completion screenshot



2. Introduction

This module covered the fundamentals of the Domain Name System (DNS), including domain hierarchy, common DNS record types, and the process used to resolve domain names into IP addresses. I learned how components such as TLDs, subdomains, A, AAAA, CNAME, MX, and TXT records work together to provide information about domains and their associated services. This is important to a security analyst because DNS can reveal useful information about a target's infrastructure, services, and potential attack surface during reconnaissance. I expected to develop a practical understanding of DNS that would help me interpret DNS information and recognize how it can be used during security investigations.

3. The work

3.1 What is DNS?

What this section covered

This section introduced the Domain Name System (DNS) and how it translates human-readable domain names into IP addresses that computers use to communicate over networks. I learned that DNS makes accessing websites more convenient by allowing users to remember names such as tryhackme.com instead of numerical IP addresses. The key takeaway for me was understanding how DNS acts as an essential link between users and the underlying network addresses of internet services.

Question

What does DNS stand for?

Answer

Domain Name System

How you got there

The section explained that DNS is an abbreviation for Domain Name System. I used the explanation provided in the module, which described DNS as the system that translates human-readable domain names into IP addresses.

3.2 Domain Hierarchy

What this section covered

This section introduced the structure of the DNS domain hierarchy, including top-level domains (TLDs), second-level domains, and subdomains. I learned how these different parts combine to form complete domain names and how subdomains can be used to organize different services or areas within a domain. The key takeaway for me was understanding how domain structures are

organized, which is useful when identifying and investigating domains during security reconnaissance.

Question

What is the maximum length of a subdomain?

Answer

63

How you got there

The module explained that a subdomain has a maximum length of 63 characters. I used the information provided in the Domain Hierarchy section, which stated that subdomain names follow the same length restriction as second-level domains. Therefore, the maximum length of a subdomain is 63 characters.

Question

Which of the following characters cannot be used in a subdomain (3 b _ -)?

Answer

_

How you got there

The module explained that subdomains can only contain letters (a-z), numbers (0-9), and hyphens (-). Since the underscore (_) was not included among the permitted characters, it cannot be used in a standard subdomain name. Therefore, the correct answer is _.

Question

What is the maximum length of a domain name?

Answer

253

How you got there

The module explained that although individual subdomains have a limit of 63 characters, the complete domain name can be up to 253 characters long.

Question

What type of TLD is .co.uk?

Answer

ccTLD

How you got there

The module explained that ccTLDs (Country Code Top-Level Domains) are associated with geographical locations or countries. The .uk part of .co.uk represents the United Kingdom, so .co.uk is classified as a ccTLD. Therefore, the correct answer is ccTLD.

3.3 Record Types

What this section covered

This section introduced common DNS record types and the different information they provide about a domain. I learned how A and AAAA records map domains to IPv4 and IPv6 addresses, while CNAME, MX, and TXT records provide information about aliases, mail servers, and domain verification or email security configurations. The key takeaway for me was understanding that DNS records can reveal useful information about a domain's infrastructure and services, making them valuable during security reconnaissance.

Question

What type of record would be used to advise where to send email?

Answer

MX

How you got there

The module explained that MX (Mail Exchange) records specify the mail servers responsible for handling email for a domain.

Question

What type of record handles IPv6 addresses?

Answer

AAAA

How you got there

The module explained that AAAA records are used to resolve domain names to IPv6 addresses. I compared this with A records, which are used for IPv4 addresses, and identified AAAA as the record specifically associated with IPv6.

3.4 Making a Request

What this section covered

This section explained the process that takes place when a device makes a DNS request to find the IP address associated with a domain name. I learned how the request can be resolved through local caches, recursive DNS servers, root servers, TLD servers, and authoritative DNS servers, with caching and TTL values helping to improve efficiency. The key takeaway for me was understanding the complete DNS resolution process and how each stage can provide useful information when analyzing network activity or investigating DNS-related security issues.

Question

What field specifies how long a DNS record should be cached for?

Answer

TTL

How you got there

The module explained that DNS records contain a TTL (Time To Live) value, which is represented in seconds and determines how long the DNS response should be stored in a local cache.

Question

What type of DNS Server is usually provided by your ISP?

Answer

recursive

How you got there

The module explained that a Recursive DNS Server is usually provided by an Internet Service Provider (ISP). It checks its own cache first and, if the requested information is not available, queries other DNS servers to find the correct answer.

Question

What type of server holds all the records for a domain?

Answer

authoritative

How you got there

The module explained that an Authoritative DNS Server is responsible for storing the DNS records for a particular domain. It provides the official DNS information for that domain, including records such as CNAME records.

3.5 Practical

What this section covered

This section involved building requests that can be used to make DNS requests and obtain useful records such as TXT records using the command line while getting help from the web interface.

I took from this section how important it is to master these commands, as they can be used during reconnaissance to obtain useful information about a target system.

Question

What is the CNAME of shop.website.thm?

Answer

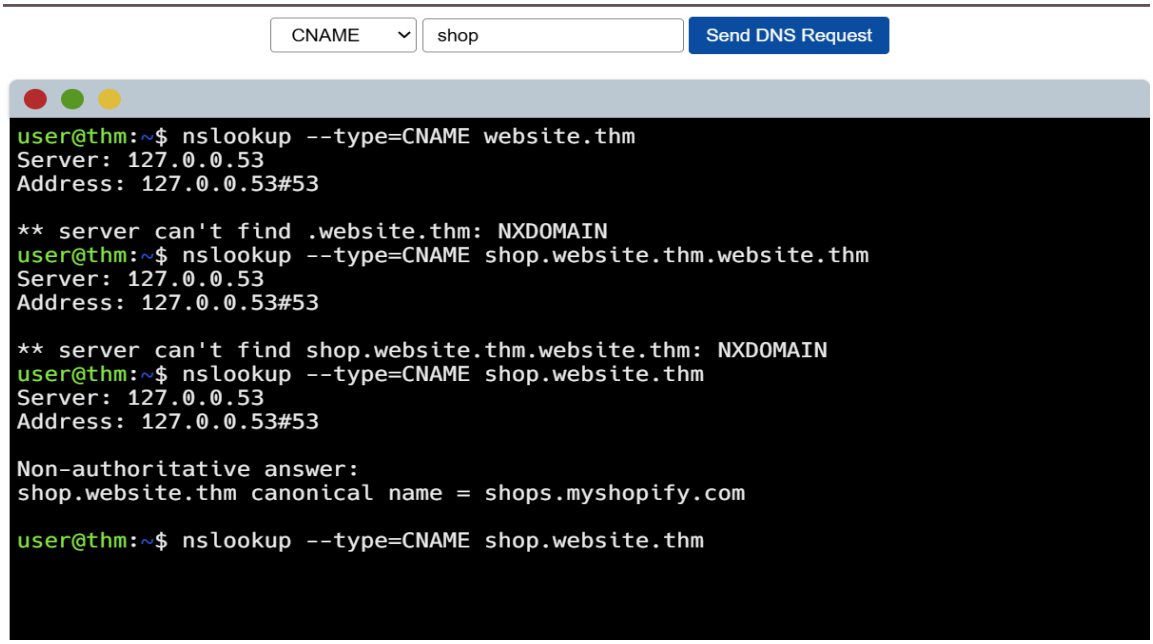
shops.myshopify.com

How you got there

In the web interface above the terminal, I selected CNAME from the DNS Type dropdown menu and entered shop in the subdomain field. After clicking Send DNS Request, the tool automatically executed the corresponding DNS lookup, nslookup --type=CNAME shop.website.thm, and returned the canonical name mapping in the terminal.

Evidence

The screenshot shows the web form configured with the CNAME record type and the shop subdomain, together with the resulting terminal output



It provides evidence that shop.website.thm points to the canonical name shops.myshopify.com.

Question

What is the value of the TXT record of website.thm?

Answer

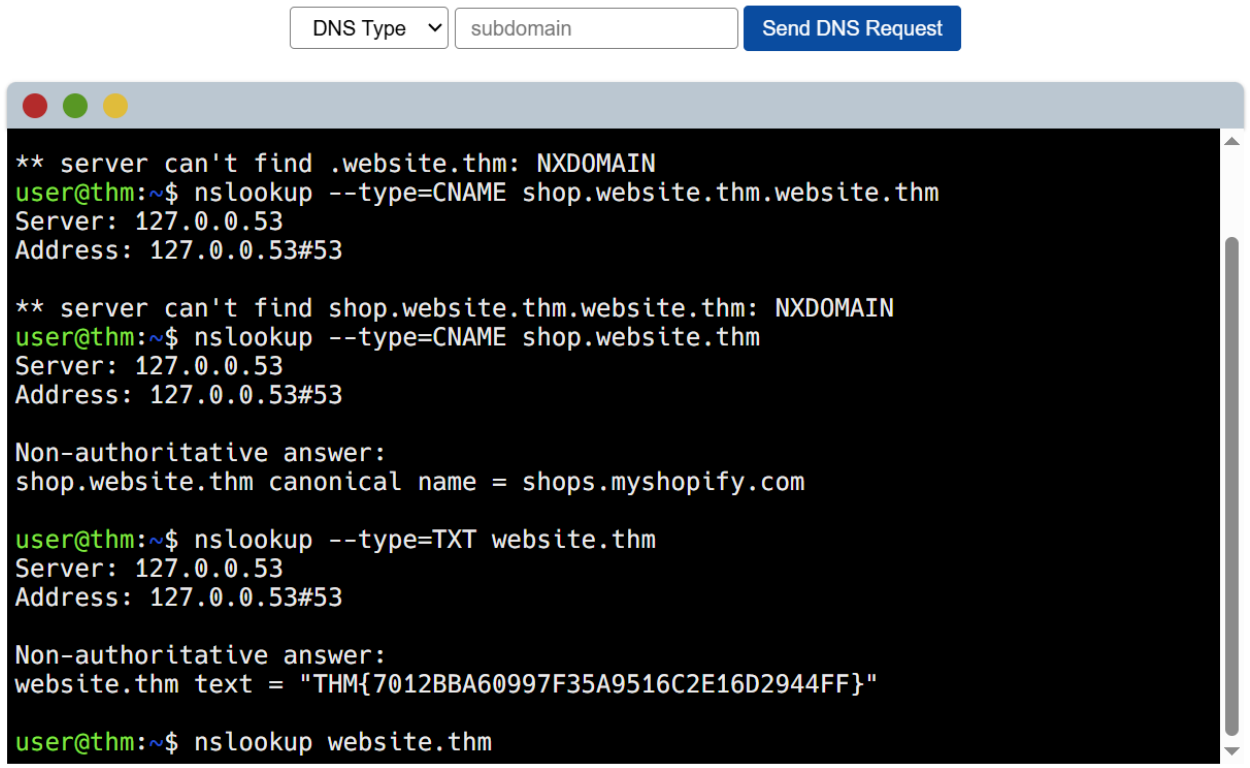
THM{7012BBA60997F35A9516C2E16D2944FF

How you got there

In the web interface, I selected TXT from the DNS Type dropdown menu and left the subdomain field blank to query the apex domain directly. After clicking Send DNS Request, the tool executed nslookup --type=TXT website.thm and displayed the domain's TXT record value in the terminal.

Evidence

The screenshot shows the web interface configured for a TXT record lookup and the resulting terminal output containing the returned TXT string.



It provides evidence that the TXT record for website.thm is THM{7012BBA60997F35A9516C2E16D2944FF}.

Question

What is the numerical priority value for the MX record?

Answer

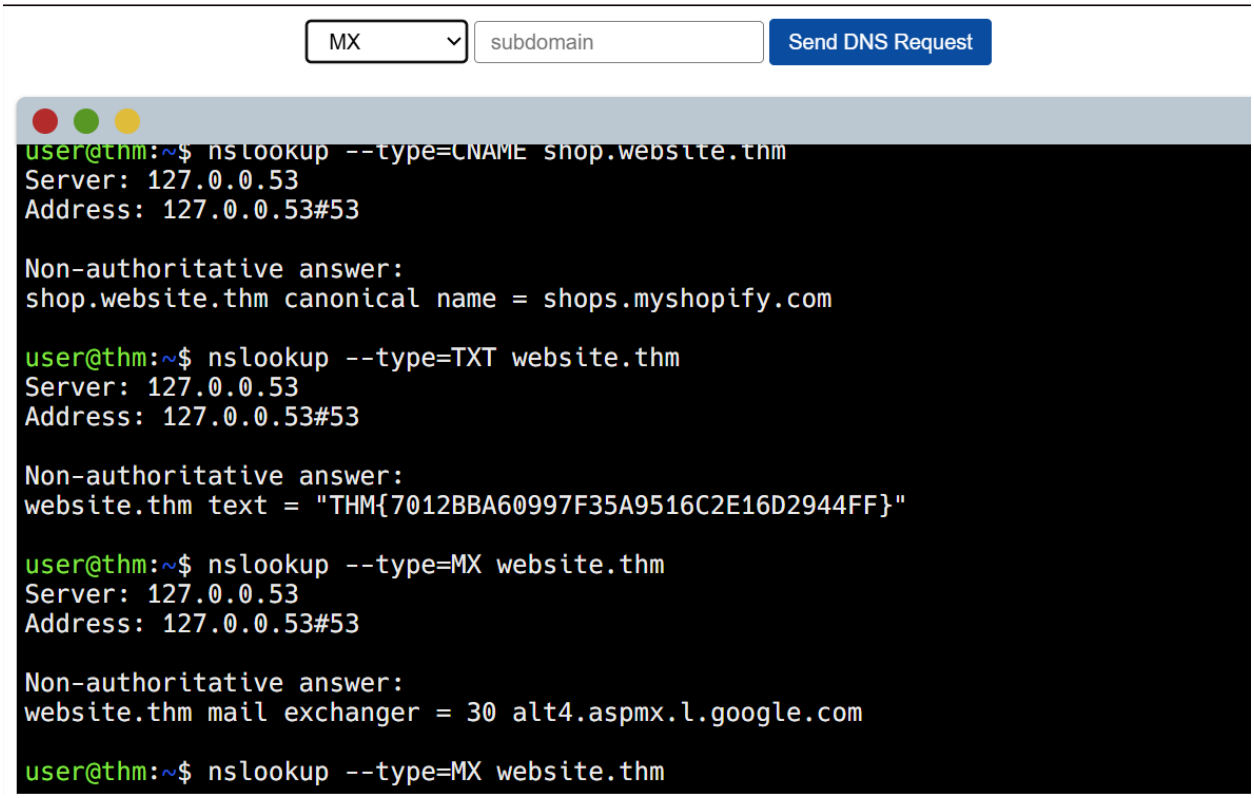
30

How you got there

I selected MX from the DNS Type dropdown menu and left the subdomain field blank before clicking Send DNS Request. The application executed nslookup --type=MX website.thm, which displayed the mail exchange server together with its numerical preference value in the terminal.

Evidence

The screenshot shows the MX record selected in the web interface and the resulting mail exchanger information in the terminal.



It provides evidence that the numerical priority value assigned to the website.thm MX record is 30.

Question

What is the IP address for the A record of www.website.thm?

Answer

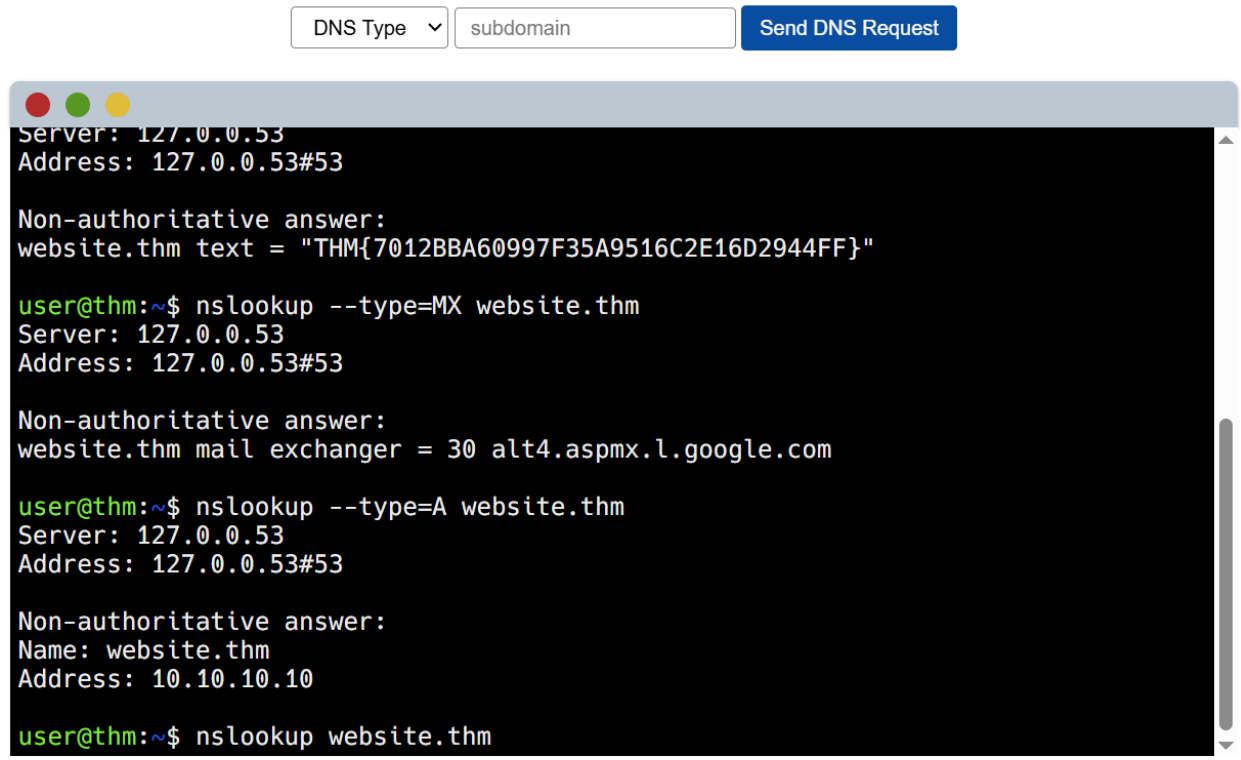
10.10.10.10

How you got there

In the web interface, I selected A from the DNS Type dropdown menu and submitted the request using Send DNS Request without specifying a subdomain. The tool executed nslookup --type=A website.thm, which returned the corresponding IPv4 address in the terminal.

Evidence

The screenshot shows the completed DNS lookup and the resulting A record resolution in the terminal.



It provides evidence that the A record for the target domain resolves to 10.10.10.10.

4. Challenges and how you resolved them

This section was not particularly challenging because the web interface provided clear guidance on how to perform the DNS queries. I was able to select the required DNS record type, enter the subdomain where necessary, and send the request without having to construct the commands manually.

The main thing I needed to do was understand the information returned in the terminal and identify the specific value required for each question. The practical interface made this easier by automatically generating the appropriate nslookup command and displaying the results.

5. Conclusion

This section improved my ability to perform practical DNS lookups and interpret the information returned by different DNS record types. I can now use DNS queries to identify information such as IP addresses, mail servers, CNAME mappings, TXT records, and MX priorities, rather than only understanding these concepts theoretically. The practical exercise also helped me see how DNS information can be useful during reconnaissance by revealing details about a target's

services and infrastructure. In future exercises, I would aim to perform more of the queries manually from the terminal to strengthen my command-line skills. Overall, this knowledge is useful to a security analyst because DNS enumeration can provide valuable information when investigating or assessing a target network.